

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

10 answers · Rationale-rich · Teach from doc

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Q1**D**

D. $f(x) = 2x - 5$

Choice D is correct. An equation defining the linear function f can be written in the form $fx = mx + b$, where m is the slope and b is the y -intercept of the graph of $y = fx$ in the xy -plane. It's given that the graph of $y = fx$ has a slope of 2. Therefore, $m = 2$. It's also given that the graph of $y = fx$ has a y -intercept at 0, -5. Therefore, $b = -5$. Substituting 2 for m and -5 for b in the equation $fx = mx + b$ yields $fx = 2x - 5$. Thus, the equation that defines f is $fx = 2x - 5$.

Choice A is incorrect. For this function, the graph of $y = fx$ in the xy -plane has a slope of $\frac{1}{2}$, not 2.

Choice B is incorrect. For this function, the graph of $y = fx$ in the xy -plane has a slope of $-\frac{1}{2}$, not 2.

Choice C is incorrect. For this function, the graph of $y = fx$ in the xy -plane has a slope of -2, not 2.

Q2

A

A.

x	y
0	4
1	5
2	6

Choice A is correct. Substituting 0 for x into the given equation yields $y = 0 + 4$, or $y = 4$. Therefore, when $x = 0$, the corresponding value of y for the given equation is 4. Substituting 1 for x into the given equation yields $y = 1 + 4$, or $y = 5$. Therefore, when $x = 1$, the corresponding value of y for the given equation is 5. Substituting 2 for x into the given equation yields $y = 2 + 4$, or $y = 6$. Therefore, when $x = 2$, the corresponding value of y for the given equation is 6. Of the choices given, only the table in choice A gives these three values of x and their corresponding values of y for the given equation.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 6$.

Choice C is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 2$.

Choice D is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = x$.

Q3

A

A. 7

Choice A is correct. Applying the distributive property on the left-hand side of the given equation yields $3x + 5x + 20 = 76$, or $8x + 20 = 76$. Subtracting 20 from each side of this equation yields $8x = 56$. Dividing each side of this equation by 8 yields $x = 7$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the solution to the equation $x + 4 = 76$, not $3x + 5x + 4 = 76$.

Q4**B****B.** 38

Choice B is correct. It's given that function f is defined by $fx = 80 - 6x$. The value of $f7$ can be found by substituting 7 for x in the given function, which yields $f7 = 80 - 67$, or $f7 = 80 - 42$, which is equivalent to $f7 = 38$. Therefore, the value of $f7$ is 38.

Choice A is incorrect. This is the value of $80 - 67$, not $80 - 67$.

Choice C is incorrect. This is the value of $80 - 61$, not $80 - 67$.

Choice D is incorrect. This is the value of $80 - 6 + 7$, not $80 - 67$.

Q5**B****B.** 10

Choice B is correct. It's given that the graph shows the possible combinations of the number of pounds of tangerines, x , and the number of pounds of lemons, y , that could be purchased for \$ 18 at a certain store. If Melvin purchased lemons and 4 pounds of tangerines for a total of \$ 18, the number of pounds of lemons he purchased is represented by the y -coordinate of the point on the graph where $x = 4$. For the graph shown, when $x = 4$, $y = 10$. Therefore, if Melvin purchased lemons and 4 pounds of tangerines for a total of \$ 18, then he purchased 10 pounds of lemons.

Choice A is incorrect. This is the number of pounds of tangerines Melvin purchased if he purchased tangerines and 4 pounds of lemons for a total of \$ 18.

Choice C is incorrect. This is the number of pounds of lemons Melvin purchased if he purchased lemons and 2 pounds of tangerines for a total of \$ 18.

Choice D is incorrect. This is the number of pounds of lemons Melvin purchased if he purchased lemons and 1 pound of tangerines for a total of \$ 18.

Q6**B****B.** 8

Choice B is correct. Dividing each side of the given equation by 3 yields $x - 9 = 8$. Therefore, the value of $x - 9$ is 8.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $3x - 27$, not $x - 9$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**A****A.** -13

Choice A is correct. The value of $h-2$ can be found by substituting -2 for x in the equation defining h . Substituting -2 for x in $hx = 3x - 7$ yields $h-2 = 3-2 - 7$, or $h-2 = -13$. Therefore, the value of $h-2$ is -13.

Choice B is incorrect. This is the value of $h-1$, not $h-2$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**D**

D. $0.04x + 0.48y = 661.76$

Choice D is correct. It's given that the food truck buys forks for \$ 0.04 each. Therefore, the cost, in dollars, of x forks can be represented by the expression $0.04x$. It's also given that the food truck buys plates for \$ 0.48 each. Therefore, the cost, in dollars, of y plates can be represented by the expression $0.48y$. Since the total cost of x forks and y plates is \$ 661.76, the equation $0.04x + 0.48y = 661.76$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation in which the food truck buys forks for \$ 0.48 each and plates for \$ 0.04 each.

Q9**D**

D. 23

Choice D is correct. It's given that $3x - 8 = 7$. Adding 8 to both sides of this equation yields $3x = 15$. Adding 8 to both sides of this equation yields $3x + 8 = 23$. Therefore, the value of $3x + 8$ is 23.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not $3x + 8$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10

D

D. $y = x - 84$

Choice D is correct. It's given that the number y is 84 less than the number x . A number that's 84 less than the number x is equivalent to 84 subtracted from the number x , or $x - 84$. Therefore, the equation $y = x - 84$ represents the relationship between x and y .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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